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Abstract

Online retailing has been growing exponentially in these last few years and slowly dominating over the retailing market, taking the lead over the classical Brick and Mortar stores. As a result, the Brick and Mortar retailers are finding it more difficult to maintain their business since the competition is growing and since the costs of preserving their physical location are becoming more and more expensive. So, in most of the times, switching from a Brick-and-Mortar store to an online platform has become more profitable.

In this internship, we are going to study the decision of a Brick-and-Mortar retailer to switch to an online platform by choosing the optimal time of switching. This study will be conducted by making this problem a stopping time problem and seeing the different parameters that will affect this decision.

Keywords : Stopping Time Problem, Howard Algorithm, Dynamic Programming, Variational Inequalities, Optimal Control

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General Introduction

The retail industry is undergoing rapid transformation. Online sales are growing at a faster rate than overall retail sales. In recent years, the demise of huge chains has garnered a lot of attention. All of this movement spurred feverish headlines about a "retail apocalypse,". Despite the fact that the number of store closures set records in 2017 and 2018, more than five times as many retailers want to establish more stores in 2019 as they did in 2018.

The major difficulties facing the brick-and-mortar retail business are: The cost of goods (which is an increasing problem especially after the pandemic and the geopolitical crisis we're living in), cost of labor (Inflation in the employee market is a concern for retailers, as shifting economic realities in many regions of the world have driven up base pay.

Along with the fluctuating cost of goods, merchants must also contend with the fluctuating cost of labour, which can be tough to afford and move quickly, leaving employers scrambling) and last but not least, the competition with online retail which is the biggest hurdle for brick and mortar retail. Online retail has been progressively snatching sales from brick-and-mortar for years. As a consequence, many of the brick-and-mortar stores are choosing to adopt online retailing as an alternative to sell their goods

In this internship, we are going to study the optimal time for a brick and mortar retailer to switch to online retailing. This work is going to be constituted of 4 major parts:

- **General Framework:** In this chapter we are going to present the host company, state the problematic and motivation behind this work and look at the previous similar works.
- **Theoretical Background:** This chapter is going to present the different theoretical tools that we used in this internship and explain them.
- **Model and added value:** In this chapter, we will build our model and state the

different changes and modifications that we made.

- **Implementation and results:** This final chapter is going to be dedicated to the implementation and results of this project.

Chapter 1

General Framework

1.1 Introduction

In this first chapter, we will introduce the host organization in which this internship was held, we will also present the motivation behind this work and our objectives. And finally we will have a look on the previous works.

1.2 Host Organization

GERAD (Group for Research in Decision Analysis) is an inter-university research center created in 1979 and brings together specialists in data and decision sciences, computer scientists, applied mathematicians and mathematical engineers, mainly from HEC Montréal, Polytechnique Montréal, McGill University and the Université du Québec à Montréal. The team is composed of more than 70 researchers from university departments in Quebec, more than 30 associate researchers from various institutions or industry, nearly 20 research professionals, and nearly 500 students at the master's, doctoral and post-doctoral levels.



Figure 1.1: GERAD logo

I have worked in GERAD with a research team led by Pr. Georges Zaccour and with the help of the Postdoc Utsav Sadana. The internship was held for 4 months.

1.3 Work motivation

E-commerce marketplaces are expanding at a rapid pace. Retail e-commerce sales globally were 2.3 trillion US dollars in 2017, with e-commerce revenues expected to reach 4.891 trillion US dollars by 2021. Traditional markets are only predicted to rise by 2% throughout the same period. Because of internet merchants' capacity to provide cheaper pricing and improved efficiency, brick and mortar stores are failing. By connecting physical and online services, many bigger shops are able to retain a presence both physically and online.

E-commerce enables users to bypass geographical constraints and purchase things at any time and from any location. Online and conventional marketplaces use distinct business tactics. Traditional stores have lower product assortments due to shelf space constraints, but internet merchants sometimes maintain no inventory and transfer consumer orders straight to the manufacturer. Pricing tactics vary between conventional and internet shops. Traditional merchants base their pricing on shop traffic and inventory costs. Prices for online shops are determined by the speed of delivery.

E-commerce has been recognized as a key contributor to the demise of large U.S. shops, a phenomenon sometimes referred to as the "retail apocalypse". The emergence of e-commerce platforms such as Amazon has made it more difficult for conventional retailers to draw people to their storefronts and compelled businesses to alter their sales techniques. Many businesses have resorted to sales campaigns and expanded internet efforts to attract customers while closing physical stores. The trend has compelled several brick-and-mortar companies to close their stores. [1]

That is why, in this work, we are going to study the strategy of a brick and mortar retailer

of closing his store and switching to an online platform, by determining the optimal time for making this decision and whether it is beneficial for his business or not, based on his market popularity and the evolution of technology of data usage.

1.4 Literature review

Plenty of previous works have studied similar problems and used similar tools that we will employ in this work.

Existing literature, such as [2] consider similar innovation problems, but they do not take into account the important feature of declining demand for the existing product.

The more recent work [3], on which we based our model, takes into consideration the declining nature of the market and presents a rigorous mathematical explanation and an analytical function that we are going to study more in detail. However, they do not present any numerical results or numerical approach to solve such problems.

Other works like [4], who studied a two player nonzero sum stochastic impulsive game , or [5], who gave results for various problems such as American option pricing, merton's portfolio problem with a compact control set or the double obstacle problem, used a policy iteration algorithm similar to the one that we are using in this work.

1.5 Conclusion

This first chapter was dedicated to present the general framework of this internship by introducing the hosting organization GERAD, the different facts that encourage and motivates us to do this study and finally getting an idea of the previous works that helped us to do this work and attain results.

Chapter 2

Theoretical Background

2.1 Introduction

In order to solve such a problem, it is imperative for our work to be backed with a set of theoretical tools. These tools, that are mainly related to dynamic programming, optimal control theory and optimization, are going to be the building bricks to attain the desired results and solving the brick and mortar or online platform problem.

In this chapter, we are going to get a general idea of the theoretical knowledge that we will need in this project.

2.2 Optimal Control Theory

2.2.1 Definition

Control Theory is concerned with controllable systems, that is, systems whose development may be altered by an external actor. In this section, we will look into control systems, which may be characterised as a system of differential equations dependent on specific parameters $u \in U \subseteq \mathbb{R}^m$

$$\dot{x} = f(x, u)$$

Where $x \in \mathbb{R}^n$. The trajectory depends on the initial point x_0 and the control parameter u and U is the set of admissible controls.

In this work, we are going to be interested in the open loop control problems where the the

control u is a function of time t .

The first issue is determining the set of positions that can be reached from x_0 using open loop controls. This is sometimes referred to as the controllability issue. If controllability to a final point x_1 is provided, one might endeavour to achieve x_1 maximizing the profit function, creating an Optimal Control Problem:

$$\max J(x, u, t_0) = \max \int_0^{t_1} I(x(t), u(t))dt + \phi(x(t_1)), \quad x(0) = x_0, x(t_1) = x_1$$

Where $I : \mathbb{R}^n \times U \rightarrow \mathbb{R}$ is the running profit and $\phi(x(t_1)) = J(x, u, t_1)$ is the profit-to-go function.

2.2.2 Pontryagin's Maximum Principle

The Pontryagin Maximum Principle is a set of conditions that must be satisfied by optimal control problems, a class of optimization problems with dynamic constraints. It unites a number of classically required requirements from the calculus of variations. This page presents an overview of the Maximum Principle, including its smooth and non-smooth forms. Time-optimal control problems are an illustration of the use of this principle.

Let $\lambda(t)$ be the time varying Lagrangian multiplier vector. The Hamiltonian H is defined as

$$H(x(t), u(t), \lambda(t), t) = \lambda^T(t)f(x(t), u(t)) + I(x(t), u(t))$$

Where λ^T is the transpose of λ

The Pontryagin's Maximum Principle the optimal control (x^*, u^*, λ^*) must satisfy the following conditions:

1. $H(x^*(t), u^*(t), \lambda^*(t), t) \geq H(x^*(t), u, \lambda^*(t), t), \forall u \in U$
2. $-\dot{\lambda}^T(t) = H_x(x^*(t), u^*(t), \lambda(t), t) = \lambda^T(t)f_x(x^*(t), u^*(t)) + I_x(x^*(t), u^*(t))$
3. $\lambda^T(t_1) = \phi_x(x(t_1))$
4. $\phi_{t_1}(x(t_1)) + H(t_1) = 0$

Condition (4) only applies if $x(t_1)$ is not fixed. If it is fixed, then this condition is unnecessary for an optimal solution.

$(\cdot)_x$ is the gradient relative to x

2.3 Value Function

The value function of an optimization problem finds the value obtained by the objective function at a solution while just relying on the problem parameters. The value function in a controlled dynamical system reflects the optimal payoff of the system across the interval $[t, t_1]$ when the system is initiated at the time- t state variable $X_t = x$. If the objective function represents a cost to be minimised, the value function may be understood as the cost to complete the optimal program, and is therefore known as the "cost-to-go function." The value function is essentially analogous to the indirect utility function in an economic setting, where the objective function generally expresses utility.

The value function in an optimal control issue is defined as the supremum of the objective function over the set of admissible controls. Given $(t_0, x_0) \in [0, t_1] \times \mathbb{R}^d$, a common optimum control problem is to solve.

$$\max J(t_0, x_0; u) = \int_{t_0}^{t_1} I(t, x(t), u(t))dt + \phi(x(t_1)), \quad \dot{x} = f(x(t), u(t))$$

Thus, the objective function V is defined as:

$$V(t, x(t)) = \max_{u \in U} \int_t^{t_1} I(\tau, x(\tau), u(\tau))d\tau + \phi(x(t_1))$$

With $V(t_1, x(t_1)) = \phi(x(t_1))$

In simpler words, the value function describes "how good" is a state or a state-action pair for the studied system or agent based on the expected return of this state or state-action pair.

Value functions are very much used in reinforcement learning and is a fundamental concept in this field.

2.4 Hamilton-Jacobi-Bellman equation

2.4.1 Definition

The Hamilton–Jacobi–Bellman (HJB) equation provides a necessary and sufficient condition for optimality of a control with regard to a loss function in optimal control theory. It is a nonlinear partial differential equation in the value function in general, which implies that its solution is the value function itself. Once this solution is found, it may be utilised to get the optimal control by maximising (or minimising) the Hamiltonian involved in the HJB equation.

The equation is the following

$$-\frac{\partial V(x, t)}{\partial t} = \max_{u \in U} \left\{ f(x, u) \cdot \frac{\partial V(x, t)}{\partial x} + I(x(t), u(t), t) \right\} \quad (2.1)$$

2.4.2 Deriving the HJB equation

To derive this equation we must first look at the dynamic programming principle. Dynamic programming is a mathematical optimization technique as well as a computer programming technique. Richard Bellman created the approach in the 1950s, and it has found applications in a wide range of industries, from aeronautical engineering to economics.

The dynamic programming principle states that an optimal policy is constituted of many sub-optimal policies.

$$V(x(t), t) = \max_{u \in U} \left\{ V(x(t + dt), t + dt) + \int_t^{t+dt} I(x(s), u(s)) ds \right\}$$

By looking at the Taylor Expansion of $V(x(t + dt), t + dt)$:

$$\begin{aligned} V(x(t + dt), t + dt) &= V(x(t), t) + \frac{\partial V(x, t)}{\partial t} dt + \frac{\partial V(x, t)}{\partial x} \cdot \dot{x}(t) dt + o(dt) = \\ &= V(x(t), t) + \frac{\partial V(x, t)}{\partial t} dt + \frac{\partial V(x, t)}{\partial x} \cdot f(x(t), u(t)) dt + o(dt) \end{aligned}$$

and simplifying by $V(x(t), t)$ from both sides and dividing by dt , we can obtain equation 2.1

2.5 Time Stopping

2.5.1 Definition

The theory of optimal stopping addresses the problem of selecting a time to conduct a particular action based on sequentially observed random variables so as to maximise an expected payoff or minimise an expected cost. In the field of statistics, where the action may be to test a hypothesis or estimate a parameter, in the field of operations research, where the action may be to replace a machine, hire a secretary, reorder goods. In the field of finance where the action may be exercise an option or buy a certain asset to maximize profit, or, in our studied problem, the action is to switch from a brick-and-mortar store to an online platform..., this type of problem is often encountered.

2.5.2 Problem statement

Let $X_t \in \mathbb{R}^n$ be a state variable and $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ a continuous function. g is called the reward function.

An optimal stopping time is a stopping time τ^* such that

$$\mathbb{E}_x [g(X_{\tau^*})] = \sup_{\tau} \mathbb{E}_x [g(X_{\tau})]$$

Where $\mathbb{E}_x [g(X_t)]$ is the expected value of $g(X_t)$ given $X_0 = x$.

In fact, the reward function can be written as follows:

$$g(X_t) = \begin{cases} g_c(X_t) & \text{if } t \leq \tau \\ g_s(X_t) & \text{if } t > \tau \end{cases}$$

g_c is the reward function in the continuation region ($t \leq \tau$).

g_s is the reward function in the stopping region ($t > \tau$).

2.6 Variational Inequality

2.6.1 Definition

Stampacchia and Ficchera independently introduced variational inequalities theory, which has emerged as an interesting and fascinating branch of applied mathematics with a wide range of applications in industry, finance, economics, social, and applied sciences. Variational inequalities may be thought of as a fresh extension of variational principles, the origins of which can be traced back to Euler, Lagrange, and the Bernoulli brothers. The application of variational principles has served as a unifying factor while also playing an essential and significant part in the development of a number of different scientific subfields. The concepts and methods of variational inequalities are now being employed in a wide range of scientific fields, where they are proving to be both fruitful and novel in their respective applications. Variational inequalities have been expanded and generalised in a number of different areas utilising innovative and cutting-edge methods.

2.6.2 Problem formulation

Let K be a non empty subset of $\mathbb{R}^n$ and let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$. To solve a variational inequality is to find $x^* \in K$ such as:

$$\langle F(x^*), y - x^* \rangle \geq 0, \quad \forall y \in K$$

Where $\langle \cdot, \cdot \rangle$ is the inner product in $\mathbb{R}^n$.

This problem is denoted $VI(F, K)$.

2.6.3 Relation to optimization

The study of the relationships between variational inequalities and optimization problems is fascinating. In reality, variational inequalities are connected to a scalar optimization problem in which the objective function is a primitive of the operator causing the inequality. In other words, the above definition may be used to problems of the following type:

$$\min_{x \in K} f(x)$$

The relation between the optimization problem and the variational inequality problem has been mostly studied in the case where K is a convex subset of $\mathbb{R}^n$ and f is a defined and differentiable function on an open set containing K such as f is a primitive of F ; meaning $f' = F$.

2.7 Howard Algorithm

Howard's algorithm, also known as policy iteration, policy improvement, strategy iteration, or strategy improvement, is a broadly applicable method for sequential decision making in the face of uncertainty and an infinite horizon. The uncertainty might be caused by "nature's" randomness, the existence of a competitor, or both.

The algorithm, in its most basic form, is a mechanism for solving finite state and action space discounted Markov decision processes. Since its inception, it has been adapted to solve a wide range of problems, including undiscounted Markov decision processes, two-player recurrent concurrent mean-payoff games, two-player discounted payoff games (turn-based or concurrent), two-player parity games, simple stochastic games, and concurrent reachability games, as well as certain well-structured infinite-state versions of many of the above models. This algorithm is also used for solving variational inequalities.

In this work we are going to use the Howard Algorithm to solve non linear problems of the form:

$$\text{solve } x \in \mathbb{R}^n, \quad \min_{\alpha \in \mathcal{A}^n} (B(\alpha)x - c(\alpha)) = 0 \quad (2.2)$$

Where $\mathcal{A}$ is a nonempty compact set and $\forall \alpha \in \mathcal{A}^n$, $B(\alpha)$ is an $n \times n$ matrix and $c(\alpha) \in \mathbb{R}^n$. Since our studied problem is a time stopping problem, then we can take $\alpha \in \{0, 1\}$. The following flowchart will explain the version of the Howard Algorithm that we are going to work with in our implementations.

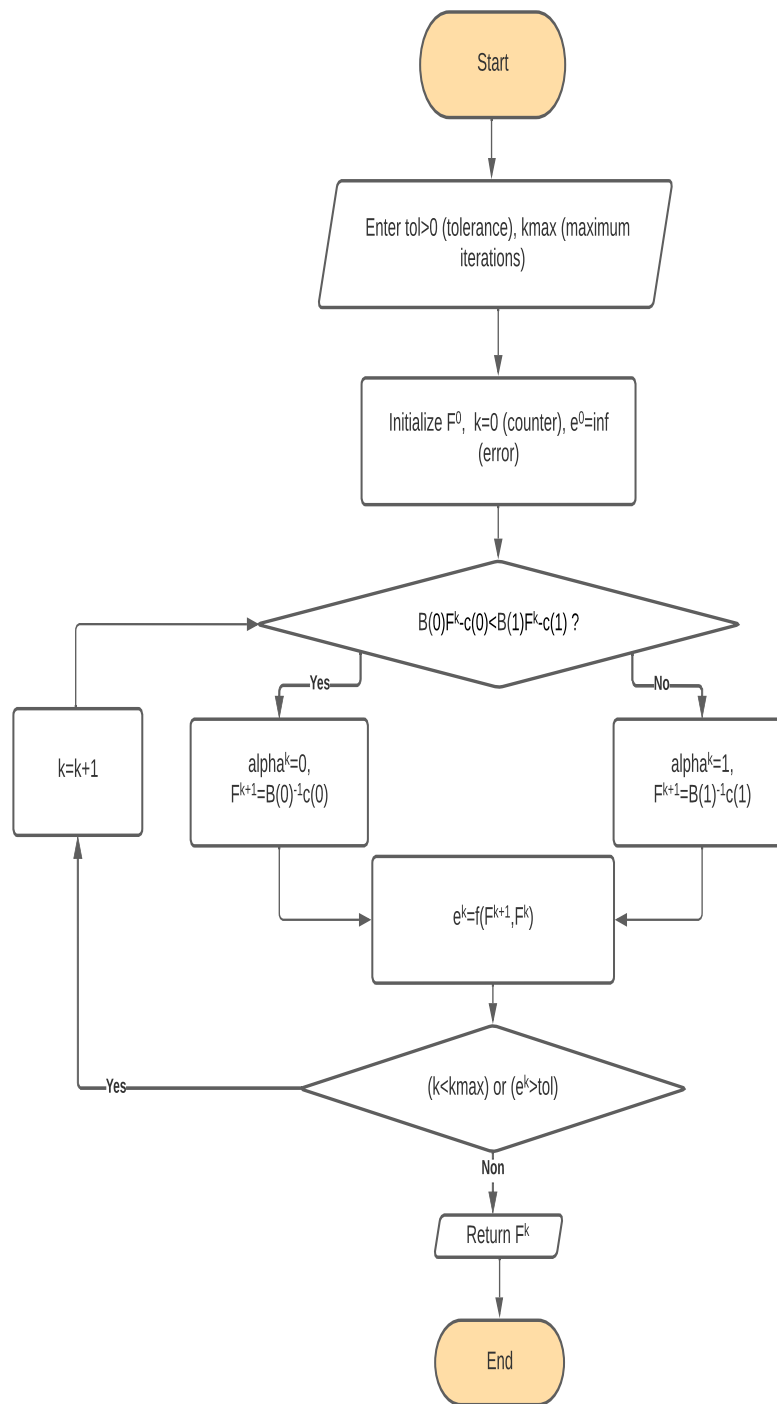


Figure 2.1: Howard Algorithm Flowchart

2.8 Conclusion

This chapter was dedicated to elucidate the theoretical knowledge necessary for solving our problem. We started by defining optimal control theory which is the basis of our work. Then we tried to explain the concept of value function which is a very strong notion used in many fields. We also presented some tools used especially in dynamic programming which are Hamilton-Jacobi-Bellman equation, stopping time problems and variational inequalities and their relation to optimization. And last but not least, we explained the briefly the Howard Algorithm which we are going to use to solve studied problem.

Chapter 3

Model and added value

3.1 Introduction

After acquiring all the theoretical tools needed to solve our problem, we are now able to look at the model of our study.

The work in this study is mostly inspired from the model elaborated in [3] with some few modifications. In this third chapter, we are going to put our focus on that model and explain it in detail. We are going to proceed by enumerating the different changes done in our study from the benchmark model.

3.2 Model

We evaluate an incumbent business that is presently doing its business in a brick-and-mortar store. In a developing economy, customers get access to superior alternatives for shopping, moving their demand away from this physical store. Furthermore, the current customer base shrinks over time, because more customers are choosing to shop from online platforms. As a result of these factors, earnings gained from the physical store market declines over time. We think that the company has been engaged in this market for some time, as a result, it has complete foresight into the existing store's popularity with customers. Hence, the profit flow of the firm at time t is deterministic and is equal to $\pi_0(X_t) = z_0 X_t$. The declining nature of the established market is captured by the process $X = \{X_t : t \geq 0\}$, where

$$dX_t = \alpha X_t dt$$

with $X_0 = x_0$ and $\alpha < 0$

The company has an incentive to refresh its buying policy since its earnings stream is on the decline. To accomplish so, it must implement an innovation strategy that includes the use of new, more sophisticated technologies in data exploitation. Innovating requires a long-term commitment of I . More important technological advancements enable a better understanding of the market. As a result of the new technology's acceptance, the firm's revenue increases as it is able to attract more customers.

The evolution of technologies through time is governed by a stochastic process exogenous to the company. A compound Poisson process, $\theta = \{\theta_t : t \geq 0\}$, determines the status of technological advancement. $\theta_t = \theta_0 + uN_t$, where $\theta_0 > 0$ represents the state of technology at the start of time, u represents the jump size, and $\{N_t, t \geq 0\}$ follows a homogeneous Poisson process with rate $\lambda > 0$. This formulation suggests that new technologies emerge at a rate of λ , and each arrival raises the degree of technology by u . It's worth noting that the process θ isn't reducing with time, which reflects the fact that technical advancement isn't diminishing.

In essence, the company has two motives to innovate. The first reason is that the existing brick-and-mortar store market profit has declined so drastically that continuing to continue doing business in this physical store is no longer economically sustainable for the business. The second reason is that, over time, new technologies of data usage have been developed that allow the business to have a better understanding of the markets than the current physical store. According to our model, the first explanation predicts a low value of X , while the second reason implies a high value of θ . We find that switching to an online platform is ideal for low values of X and high values of θ , whereas staying active in the existing store is optimal when X is high and θ is low.

The firm's goal is therefore to discover the best timing to implement new technologies. At that point, the company must determine whether to simply switch to an online platform and close the physical store, or to add a new online platform along her physical store allowing the company to do business from both of the stores. In this work we will only focus on the first option.

The firm has to choose the timing of switching to the new method of retailing. So, the company has to solve the following optimal stopping problem:

$$F(\theta, x) = \sup_{\tau} \mathbb{E} \left[\int_0^{\tau} \pi_0(X_s) e^{-rs} ds + \left\{ \int_{\tau}^{+\infty} \pi_1(\theta_{\tau}) e^{-rs} ds - I e^{-r\tau} \right\} \chi_{\{\tau < +\infty\}} \mid \theta_0 = \theta, X_0 = x \right] \quad (3.1)$$

Where

- F is the value of the firm
- τ is the investment timing
- $r > 0$ is the discount rate
- π_1 is the profit of the new online platform

In this problem formulation, the company is in a trade-off between switching to the online platform at an early time and the significance of the technological improvement. In other words, the firm has to choose between waiting for more sophisticated technology and operating in a declining market or switching at an early time with technology that can make more improvements in the future.

In case the company switches to an online platform with the current level of technology θ , it gains a profit flow $\pi_1(\theta)$. after integrating and discounting this profit flow and paying the cost of investment I . The company earns a total value of instantaneous investment.

$$V(\theta) = \frac{\pi_1(\theta)}{r} - I \quad (3.2)$$

In the [3] paper, there are a few conditions on the profit flow of the new market function π_1 for reasons of existence and uniqueness of the solution

- π_1 is a concave function
- $\pi_1(0) = 0$
- $\lim_{\theta \rightarrow +\infty} \pi_1(\theta) = \infty$

The previous conditions on π_1 imply that V is an increasing concave function which can guarantee that a unique analytic solution of the time stopping problem exists.

Solving equation (1) is equivalent to solving the following HJB equation:

$$\min \{rF(\theta, x) - [\pi_0(x) + \mathcal{L}F(\theta, x)], F(\theta, x) - V(\theta)\} = 0 \quad (3.3)$$

Where $\mathcal{L}$ is the infinitesimal generator defined by:

$$\mathcal{L}f(\theta, x) = \alpha x \frac{\partial f(\theta, x)}{\partial x} + \lambda[f(\theta + u, x) - f(\theta, x)]$$

f is continuous and differentiable in x and continuous in θ .

Let $\mathcal{S} = \{(\theta, x) \in \mathbb{R}^+ \times \mathbb{R}^+ : F(\theta, x) = V(\theta)\}$ denote the stopping region, and $\mathcal{C} = \{(\theta, x) \in \mathbb{R}^+ \times \mathbb{R}^+ : F(\theta, x) > V(\theta)\}$ denote the continuation region.

The optimal stopping time τ^* is given by the following expression

$$\tau^* = \inf\{t > 0 : (\theta_t, X_t) \notin \mathcal{C}\}$$

In other words, the optimal stopping time is the instant where continuation region is passed.

Let $b(\theta) = \frac{(r+\lambda)V(\theta) - \lambda V(\theta+u)}{z_0}$ the market popularity level that triggers investment (stopping time) for a given level of technology θ .

The analytical expression of the value function in the continuation region is given by the

following expression:

$$\begin{aligned}
 F(\theta, x) = & \left(\frac{\lambda}{r + \lambda} \right)^{n(\theta, x)} V(\theta + n(\theta, x)u) + \frac{z_0 x}{r - \alpha} \left[1 - \left(\frac{\lambda}{r + \lambda - \alpha} \right)^{n(\theta, x)} \right] \\
 & + \sum_{k=0}^{n(\theta, x)-1} \left\{ \left[\frac{x}{b(\theta + ku)} \right]^{\frac{r+\lambda}{\alpha}} z_0 b(\theta + ku) \lambda^k \times \right. \\
 & \left. \sum_{m=0}^k \frac{1}{m!(-\alpha)^m} \left[\frac{1}{(r + \lambda)^{k-m+1}} - \frac{1}{(r + \lambda - \alpha)^{k-m+1}} \right] \left[\ln \left[\frac{x}{b(\theta + ku)} \right] \right]^m \right\} \chi_{\{\bar{\theta} - ku\}}
 \end{aligned} \tag{3.4}$$

Where:

- χ_A the indicator function of set A
- $n(\theta, x)$ is the number of jumps of technology necessary until switching to an online platform, $n(\theta, x) = \left\lceil \frac{b^{-1}(x) - \theta}{u} \right\rceil$
- for $y \geq 0$, $\lceil y \rceil = \min\{n \in \mathbb{N}, n \geq y\}$

3.3 Added Value

This study is highly inspired by the work done in [3] and the model is practically the same. However there are some lacking points in this model that we are going to try to correct them in this project.

3.3.1 The inverse function

The analytical function of the continuation region presented in [3] needs the calculation of the function $n(\theta, x)$ which needs in its turn the calculation of $b^{-1}(x)$, the inverse of the function $b(x)$. The inversion of this function is highly dependent on the function the profit flow function of innovation π_1 .

In fact inverting this function when it is not linear is highly difficult since changing the parameters will completely change the form of the inverse function. The inversion is more possible when working with a linear function of π_1 . This is a big restriction for studying the

impact of the profit flow function of innovation on the decision of the retailer.

In our work, such inversions are not needed to find the value function of the retailer. This gives us a wider range of possibilities for the profit flow function to get a better insight on its effect on the optimal stopping time problem.

3.3.2 Stopping region

The results shown in [3], only propose an analytical solution for the value function in the continuation region. This leaves us with a big question mark on the nature of the stopping region and how the value function may behave in this region.

In this project, we will have a better idea on the different characteristics of the stopping region and the value function correspondent to this region, their forms, how they may be affected with the different parameters of the problem. This is highly critical to get a full view on the results of the decision of the retailer when switching to an online platform, which will help him to make the better decision.

3.3.3 Numerical results

In the paper [3], Only analytical results of the value function have been presented. The analytical function can give us an exact solution of the problem, and thanks to it we can tell the behaviour of the value function at each point of the continuous space. However, the paper lacks numerical results and does not give a visualisation of the value function to get a better understanding about it. Overseeing these numerical results, also leaves with a question on the feasibility of computing the analytical function in terms of time execution and parameters.

During the internship, it was imperative for our work to visualise the results presented in [3] to compare it with our results. It turned out that there are quite a few problems that we faced when we computed this analytical function.

First of all, the term $\left[\frac{1}{(r+\lambda)^{k-m+1}} - \frac{1}{(r+\lambda-\alpha)^{k-m+1}} \right]$ in the equation 3.4 get big values since $r+\lambda$ and $r = \lambda+\alpha$ are inferior to 1 and the exponents are relatively big ($n(\theta, x)$ is responsible for the range of the exponents). So if we do not choose our parameters carefully, we end up with infinite values in many points of the value function.

Here are a few examples of the results we get for the parameters: $\lambda = 0.1, I = 30, r = 0.45, u = 0.5, z_1 = 20, \pi_1(\theta) = z_1 x$

(θ, x)	$n(\theta, x)$	$F(\theta, x)$
(30,0)	132	$4, 54.10^{30}$
(23.5,0.15)	50	$7, 97.10^{13}$
(19,2)	10	44.59

Table 3.1: Values of the analytical function

A second thing that we noticed after this computation is that the running time of this function is quite elevated. In fact for a matrix F of size 200×30 , the running time has exceeded 1 hour 40 minutes. This time disadvantage may be resolved by using parallel computing, but for simple computations this presents a problem for us.

3.4 Conclusion

Throughout this chapter, we gave a detailed explanation of the model that we are going to work with and tried to show its different characteristics and parameters explicitly. We also showed the main differences between the benchmark work and ours and how this study will give plenty of added value.

Now let us look at the implementation of this method and the results given by it.

Chapter 4

Implementation and results

4.1 Introduction

In this chapter, we are going to give the implementation and the different techniques we used to obtain results. We will also present the obtained results and interpret them.

4.2 Implementation

Let $\mathbb{X} = [0..10]$ be an N size vector with a constant step h . X is the grid that contains all the states of the market popularity of the firm.

Let $\Theta = [0..\theta_{max}]$ be a vector with a constant step u (the jump size of the level of technology).

Θ is the grid that contains all the states of the technology levels. Let M_θ be the size of Θ

Our goal is to find value function $F(\theta, x)$ which is going to be an $M_\theta \times N$ matrix in $\mathbb{R}$.

$$F = \begin{pmatrix} F_{0,0} & F_{0,1} & \dots & F_{0,N-1} \\ F_{1,0} & F_{1,1} & \dots & F_{1,N-1} \\ \vdots & \vdots & & \vdots \\ F_{M_\theta-1,0} & F_{M_\theta-1,1} & \dots & F_{M_\theta-1,N-1} \end{pmatrix}$$

Where $F_{i,j} = F(\theta_i, x_j)$, $\forall (i, j) \in \llbracket 0, M_\theta - 2 \rrbracket \times \llbracket 0, N - 2 \rrbracket$ and $F_{M_\theta-1,N-1} = F(\theta_{max}, x_{max})$

4.2.1 Derivatives

To solve the HJB equation 3.3, we need to implement the infinitesimal generator defined in section (2.2) $\mathcal{L}$. for that we will need two operators: D_x to approximate the x derivative

$\frac{\partial f(\theta, x)}{\partial x}$, and D_θ that will give the expression $f(\theta + u, x) - f(\theta, x)$.

D_x is an $N \times N$ matrix. To find the expression of D_x , we need to look at its compact expression

$$(FD_x)_{i,j} = \frac{F_{i,j+1} - F_{i,j}}{h}$$

From the previous expression we can extract the real expression of the matrix D_x

$$D_x = \frac{1}{h} \begin{pmatrix} 0 & -1 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1 & -1 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & -1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 & -1 \\ 0 & 0 & 0 & \dots & 0 & 0 & 1 \end{pmatrix}$$

D_θ is an $M_\theta \times M_\theta$ matrix. Similarly, to find its expression we will use the following compact expression:

$$(D_\theta F)_{i,j} = F_{i,j+1} - F_{i,j}$$

Thus we can find the expression of D_θ

$$D_\theta = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ -1 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & 0 & 0 \\ 0 & 0 & 0 & \dots & -1 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{pmatrix}$$

Hence, we can compute the derivatives according to variables x and θ and we can obtain the infinitesimal generator.

4.2.2 Sylvester Equation

The problem treated in the paper [5], of the form, $\min_{\alpha \in \mathcal{A}^n} (B(\alpha)x - c(\alpha)) = 0$ where x is a one dimensional vector, can be solved by inverting the matrix $B(\alpha)$ and multiplying it by the vector $c(\alpha)$. In our problem, we are trying to solve 3.3, where we obtain an equation of the form $AY + YB = C$, where all the terms of the equation are matrices. This form of equations is defined as the Sylvester equation.

The problem here is to determine, given the matrices A, B and C , all of the potential matrices Y that obey this equation. We will proceed on the assumption that every matrix has coefficients in the complex numbers. In order for the equation to make sense, the matrices need to have the same dimensions; for instance, they may each be square matrices of the same size. However, on a more basic level, A and B need to be square matrices, and their respective widths need to be n and m . After that, Y and C should each have n rows and m columns. In our case $A = rI_{M_\theta} - \lambda D_\theta$, $B = -\alpha D_x$, $C = \pi_0$ and $Y = F$; I_n is the identity function of size n , $\forall n \in \mathbb{N}^*$

With all these information and tools, we can now implement our code.

4.2.3 Algorithm

Algorithm 1 Howard Algorithm

Set $\epsilon > 0$ (*tolerance*) and $k_{max} > 0$ (*maximum iterations*)
Initialize F^0
 $k \leftarrow 0$
 $R \leftarrow inf$
while $k < k_{max}$ and $R > \epsilon$ **do**
 For $i=0, \dots, M_\theta - 1$, $j=0, \dots, N - 1$
 $a_{i,j}^{k+1} := \begin{cases} 0 & \text{if } (rF^k(\theta, x) - [\pi_0(x) + \mathcal{L}F^k(\theta, x)])_{i,j} < (F^k(\theta, x) - V(\theta))_{i,j} \\ 1 & \text{otherwise.} \end{cases}$
 For $i=0, \dots, M_\theta - 1$, $j=0, \dots, N - 1$
 $F_{i,j}^{k+1} := \begin{cases} \text{solve the Sylvester Equation} & \text{if } a_{i,j}^{k+1} = 0 \\ (V(\theta))_{i,j} & \text{otherwise.} \end{cases}$
 $R \leftarrow \phi(F^k, F^{k+1})$
 $k \leftarrow k + 1$

$\phi(F^k, F^{k+1}) = \frac{1}{\sqrt{N \cdot M_\theta}} \sqrt{\sum_{i=0}^{M_\theta} \sum_{j=0}^{N-1} (F_{i,j}^{k+1} - F_{i,j}^k)^2}$ is the error function between F^{k+1} and F^k .

a is an $M_\theta \times N$ matrix which determines the control (either to switch to a online platform or not).

to solve the Sylvester Equation, we use the Python command *solve_sylvester* from the library *scipy.linalg*. This command takes the matrices A, B and C as arguments to find the solution Y .

4.3 Results

To present our results, we will first do a general presentation to get an insight on the results and how we can interpret them. Then, we will proceed by doing a sensitivity analysis to see how modifying the different parameters in our algorithm will affect our objective function and how we can interpret these results.

The parameters that we are going to verify are the following:

- h : The step of the grid $\mathbb{X}$.
- u : The technology jump.
- I : The cost of innovation.
- π_1 : the profit flow after innovating.

4.3.1 Overview

The results which we will present in all of this section, will be constituted of two graphics: a surface in red illustrating the value function, and a blue surface illustrating the control or the continuation and stopping regions.

Let us see the results. We are going to choose the following parameters: $h = 0.05, u = 0.5, \lambda = 0.1, I = 50, r = 0.15, \alpha = -0.5, \pi_1(\theta) = 2z_1 \ln(x + 1)$

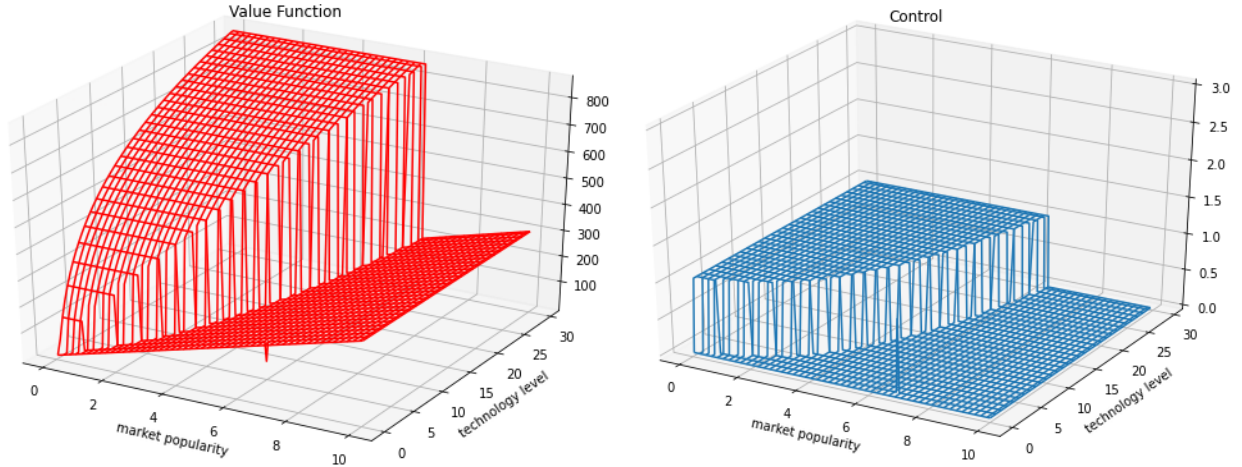


Figure 4.1: Control and Value Function

In the Control surface, we notice the existence of two regions. A continuation region where the control $\alpha = 0$. In this region, it is better for the retailer to not innovate and continue doing his business in the brick and mortar store. The second region is the stopping region where the control $\alpha = 1$. In this region, it is optimal for the retailer to innovate and switch to an online platform. We can see that the stopping happens for low values of the market popularity X and for high levels of technology: the lower the market popularity is, the higher the range of technology level where it is optimal to switch to an online platform which is very logical.

As for the Value function, we can also see two different regions. A region where the value function is linear, this corresponds to the continuation region and the function is the solution of the differential equation of 3.3. The second region has the shape of the profit flow function of innovation, it corresponds to the stopping region where the value function equals $V(\theta)$.

4.3.2 The market popularity grid step

The step of the grid has a huge role on the x derivative operator D_x . The smaller the step h is, the smoother the results will be and the closer the numerical derivative will be to the real derivative function since this value tends to zero in the analytical form.

On the other hand, getting this parameter closer to zero will increase the number of points of the matrices and thus the time to execute the algorithm will increase.

So to sum up, when choosing the value of this parameter, we are facing a trade-off between the accuracy of the model and how close it is to the real solution, and the time complexity of the algorithm.

In the sensitivity analysis for this parameter, we will not pay attention to the shape of the value function surface nor the control function surface, since we will explain them in more detail in the later sections, but rather, the distortion created in one of the regions.

To visualise the results in this phase, we chose the following parameters: $u = 0.5$, $\lambda = 0.1$, $I = 50$, $r = 0.15$, $\alpha = -0.5$, $\pi_1(\theta) = 2z_1 \ln(x + 1)$ Let us now see the effect of the change of the parameter h on our model.

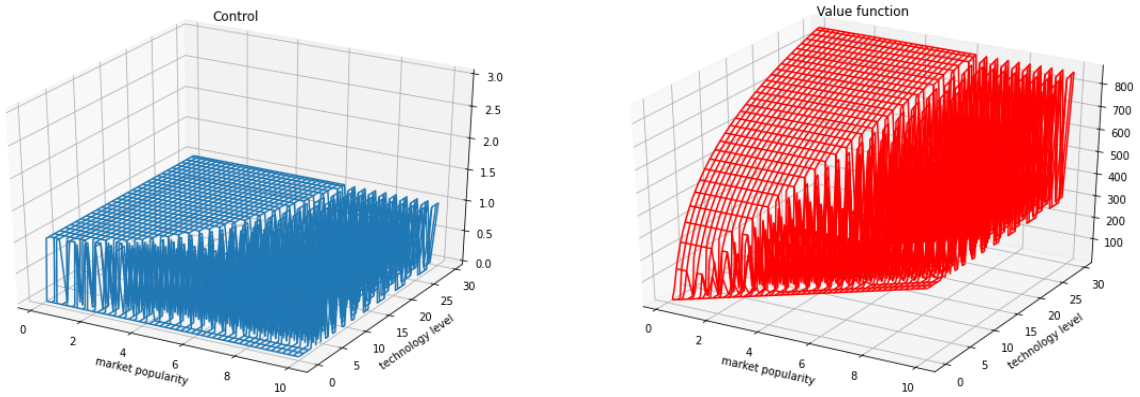
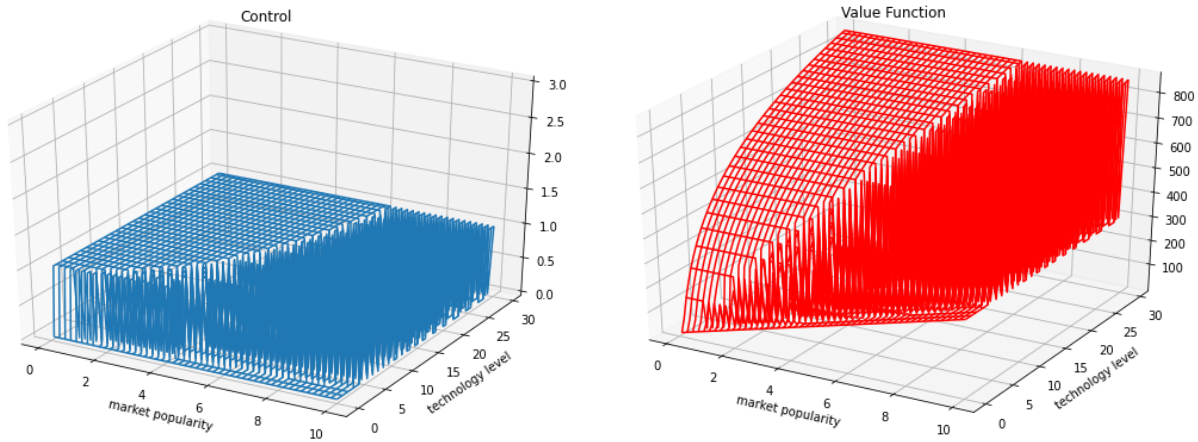
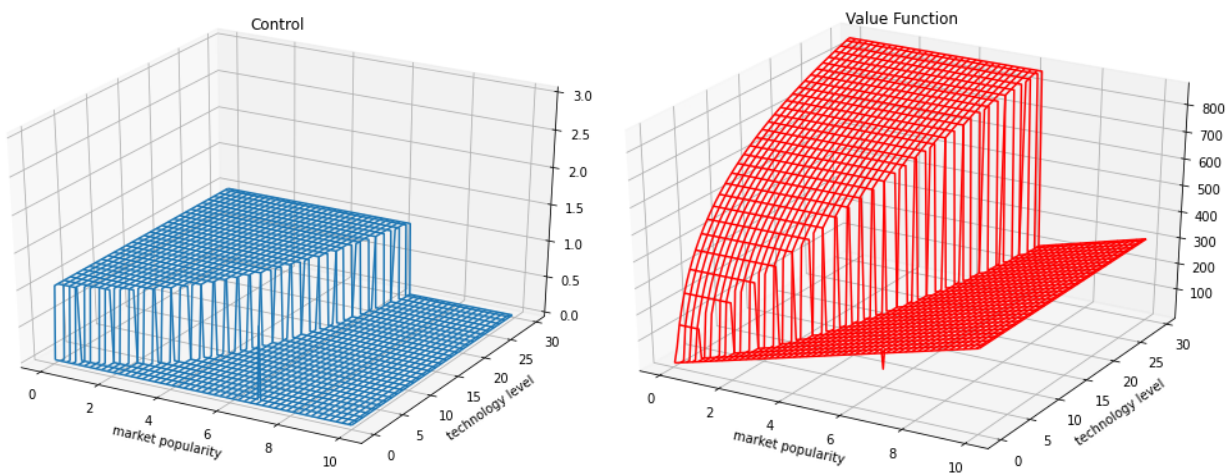
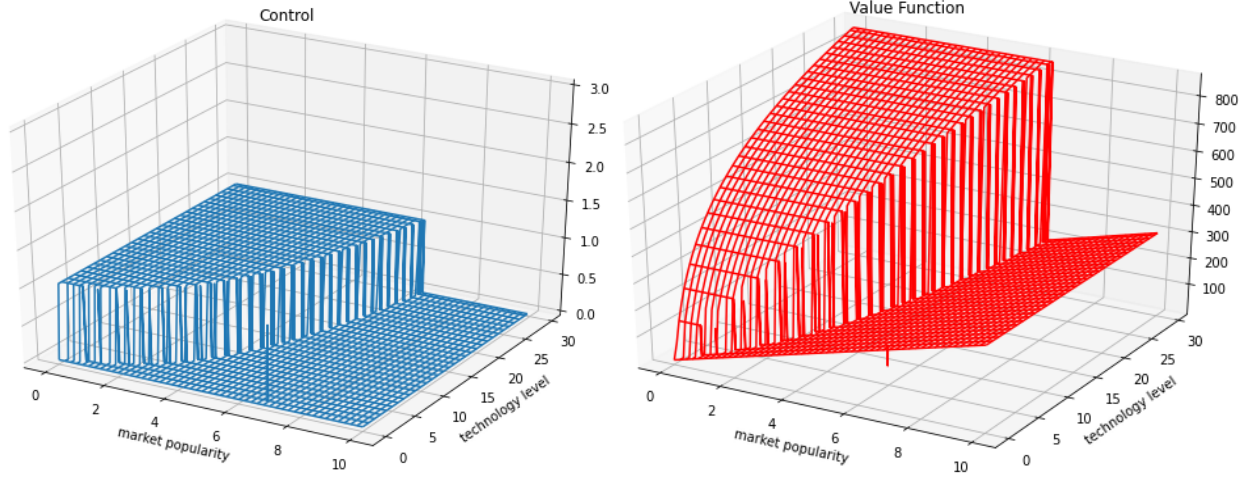


Figure 4.2: Control and Value Function for $h=0.2$

Figure 4.3: Control and Value Function for $h=0.1$ Figure 4.4: Control and Value Function for $h=0.05$

Figure 4.5: Control and Value Function for $h=0.01$

In the figures 4.2 and 4.3, for relatively bigger values of h , we can see that there is a sort of noise in the area where the control is 0. This noise is due to calculating the derivative along x and solving the Sylvester Equation using the Python solver. As for the smaller values of h , in the figures 4.4 and 4.5, we notice that we can distinguish between two different regions and that there is practically no noise in both of these regions. So we can deduce that smaller steps of h are more convenient for our computations. Now let's look at the run-time of these algorithms.

h	Number of Iterations	Run-time/iteration
0.2	40	1s
0.1	75	2s
0.05	145	7s
0.01	752	10m12s

Table 4.1: Run-time for various values of h

This table tells us about the run-time of the Howard Algorithm depending on the step h . For the first two values of h , the run-time/iteration and the number of iterations are relatively low but, as we mentioned earlier, there is noise in the surfaces. For the value $h = 0.01$, the number of iterations and the run-time iteration are quite big and it would be inconvenient, even though it gives us more accurate figures, to pursue our calculations with such a long

run-time. As for the value $h = 0.05$ the run-time is more reasonable and it gives us a much better accuracy than big values of h even though it takes longer to run.

So, from here on out, we are going to work with a value of $h = 0.05$

4.3.3 Technology jump level

When an upgrade occurs in the technology of data exploitation, it happens with a level u . This level of jump won't affect the values of θ or x since the profit in case of innovation doesn't depend on the jump u but rather on the value of the technology θ . However this level will affect the waiting time to attain a certain level of technology: for example if we are at a level of technology $\theta_0 = 5$ and we are waiting to attain a level of technology $\theta_1 = 5.5$. If the technology jump level $u = 0.5$ we will wait for one jump which occurs at a rate λ . However if the technology jump level $u = 0.1$ we will wait for five jumps to attain our desired technology level.

Computationally speaking, decreasing the jump level will increase the number of points of the matrices and hence the run-time and the number of iterations of the algorithm.

Let us take a look at the results. To visualise these results we chose the following parameters:

$$h = 0.05, \lambda = 0.1, I = 50, r = 0.15, \alpha = -0.5, \pi_1(\theta) = 2z_1 \ln(x + 1)$$

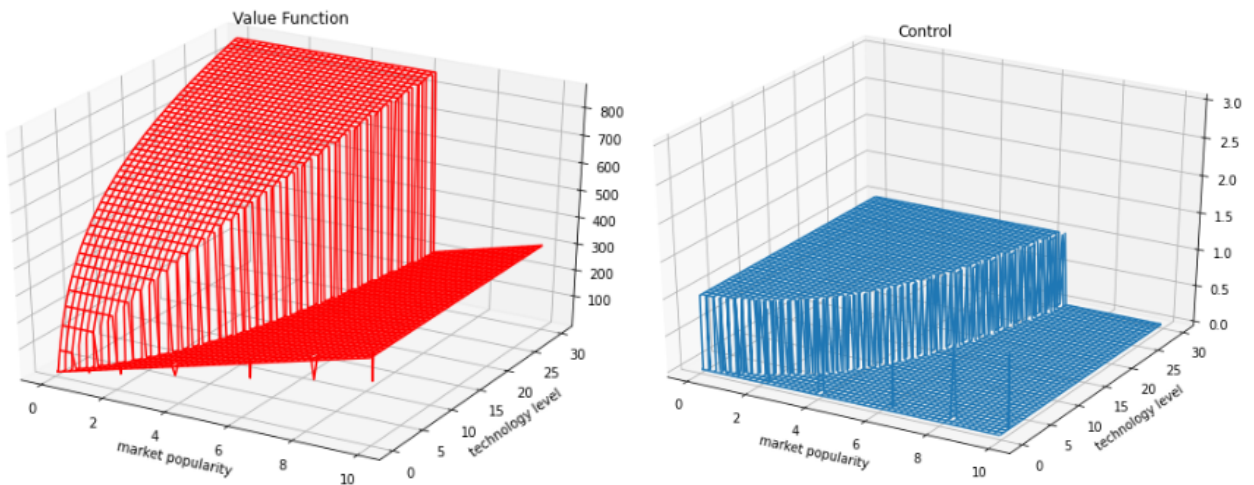
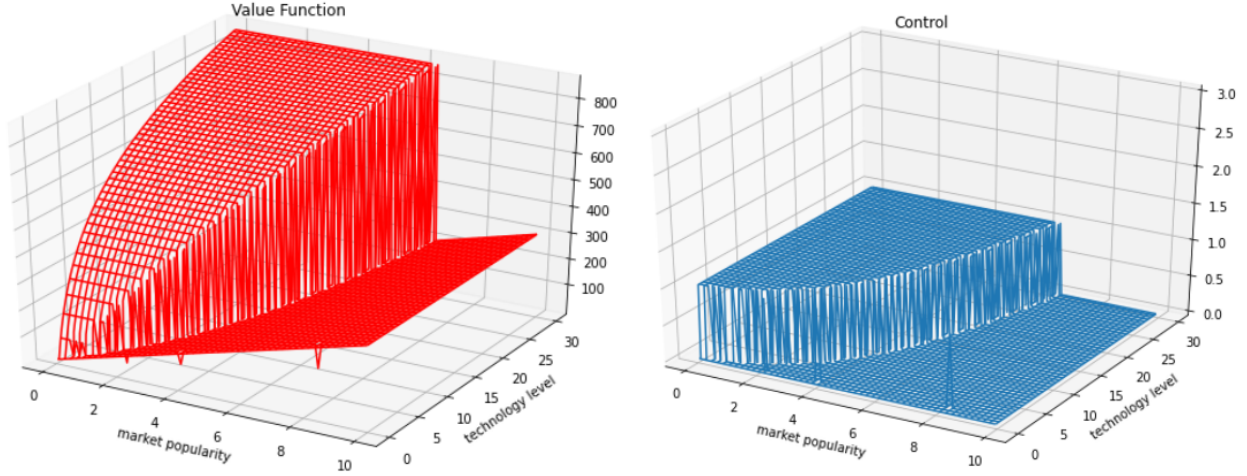
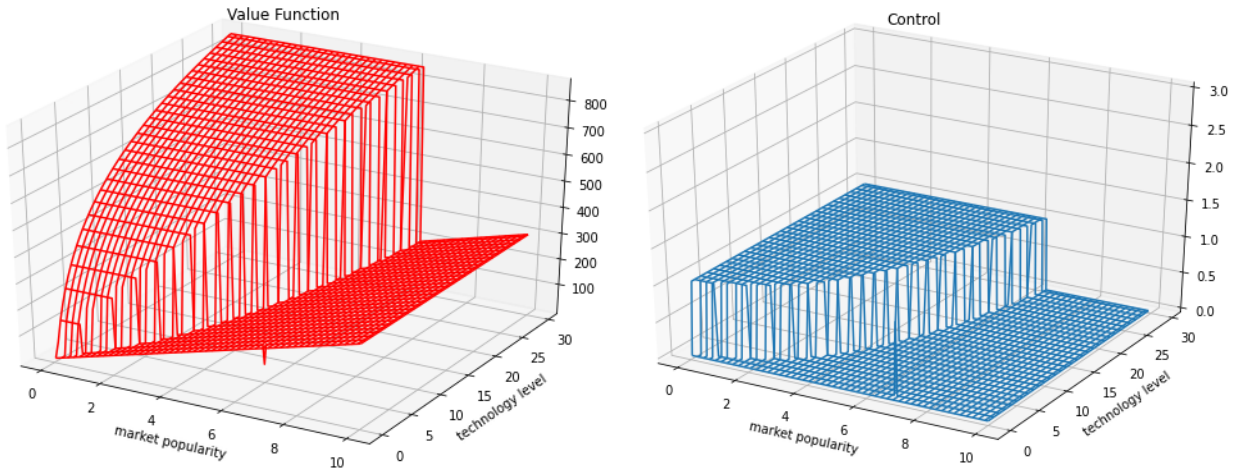


Figure 4.6: Control and Value Function for $u=0.1$

Figure 4.7: Control and Value Function for $u=0.2$ Figure 4.8: Control and Value Function for $u=0.5$

We can see that the Control surface and the Value Function surface are similar in the shape and the only difference is the density of points.

u	Number of Iterations	Run-time/iteration
0.1	248	1m5s
0.2	198	30s
0.5	145	7s

Table 4.2: Run-time for various values of u

Small values of u make the model take more time to run, and since it doesn't affect the shape of the results, it's more convenient to work with a relatively high value.

4.3.4 The cost of innovation

The cost of innovation has a big impact on the decision of switching to online retailing. In fact, if the cost of innovation I is high enough, the retailer would prefer to do the switching for higher values of technology and lower values of the market popularity and thus the surface of the smaller region decreases. A lower value of the cost I encourages the retailer to innovate at lower values of the technology level. Let us check the results.

To visualise these results we chose the following parameters: $h = 0.05, u = 0.5, \lambda = 0.1, r = 0.15, \alpha = -0.5, \pi_1(\theta) = 2z_1 \ln(x + 1)$

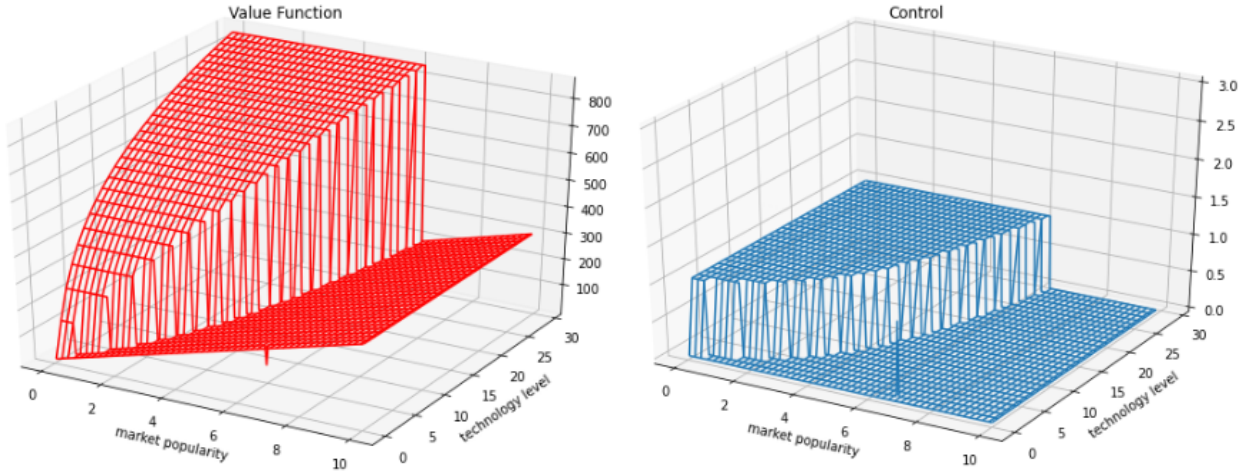
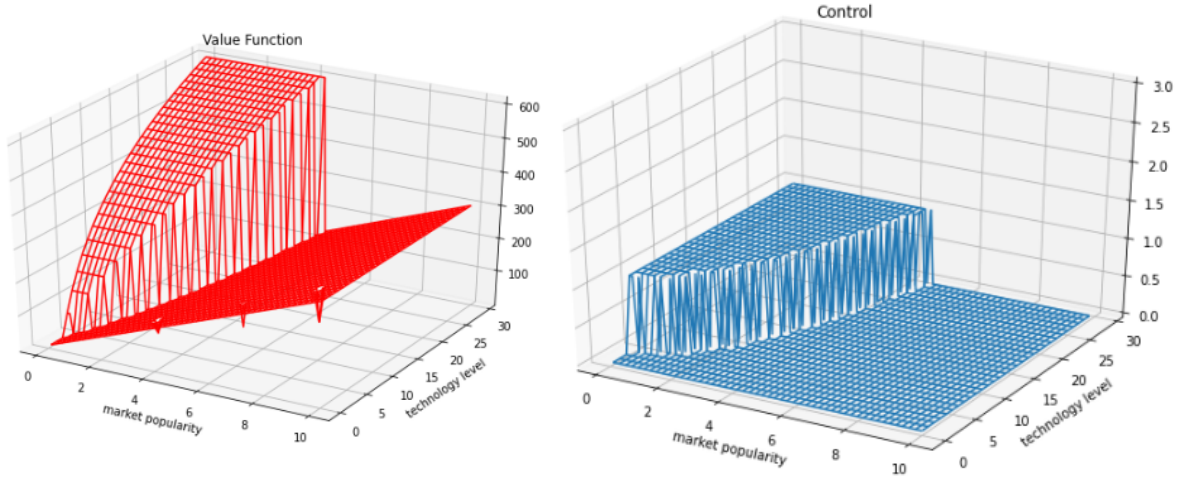
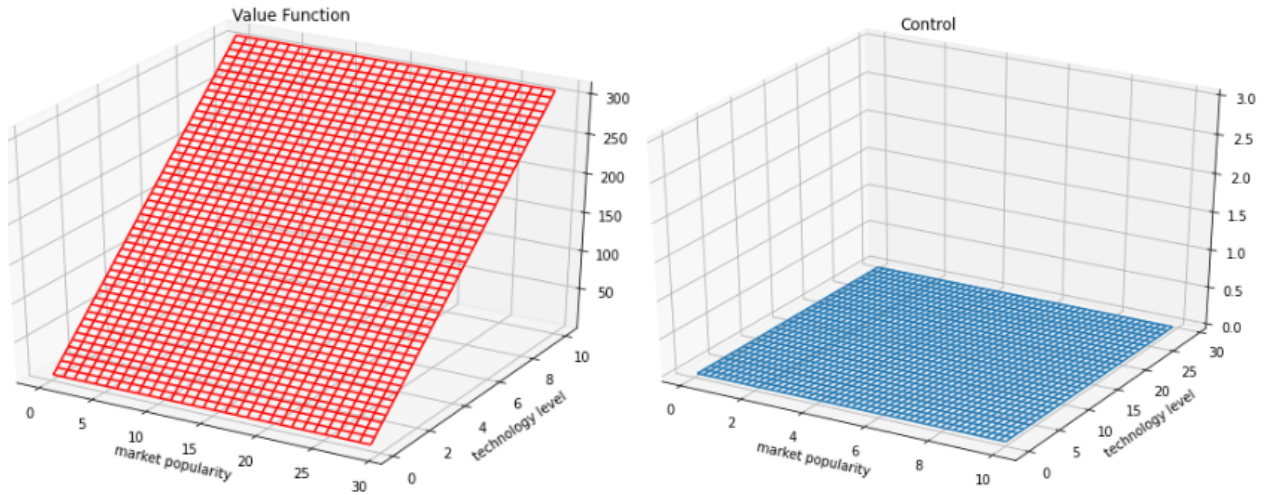


Figure 4.9: Control and Value Function for $I=50$

Figure 4.10: Control and Value Function for $I=300$ Figure 4.11: Control and Value Function for $I=1000$

We can see that the stopping region in the figure 4.9 with $I = 50$ is bigger than the stopping region in the figure 4.10 where the costs are more expensive $I = 300$. As for the figure 4.11, there is no stopping region at all since the innovation costs are much more expensive ($I = 1000$): it is no longer profitable for the retailer to invest in online retailing.

4.3.5 Profit flow function in innovation π_1

The profit flow function in innovation is also one of the factors that impacts the decision of the retailer. We can easily understand that a profit flow function that has higher values

is more profitable for the retailer to innovate. So a big profit flow function will encourage the retailer to innovate at lower values of technology and earlier in time. Let us compare the Value Function and Control surface for different functions of profit flow in innovation.

To visualise these results we chose the following parameters: $h = 0.05, u = 0.5, \lambda = 0.1, I = 50, r = 0.15, \alpha = -0.5$

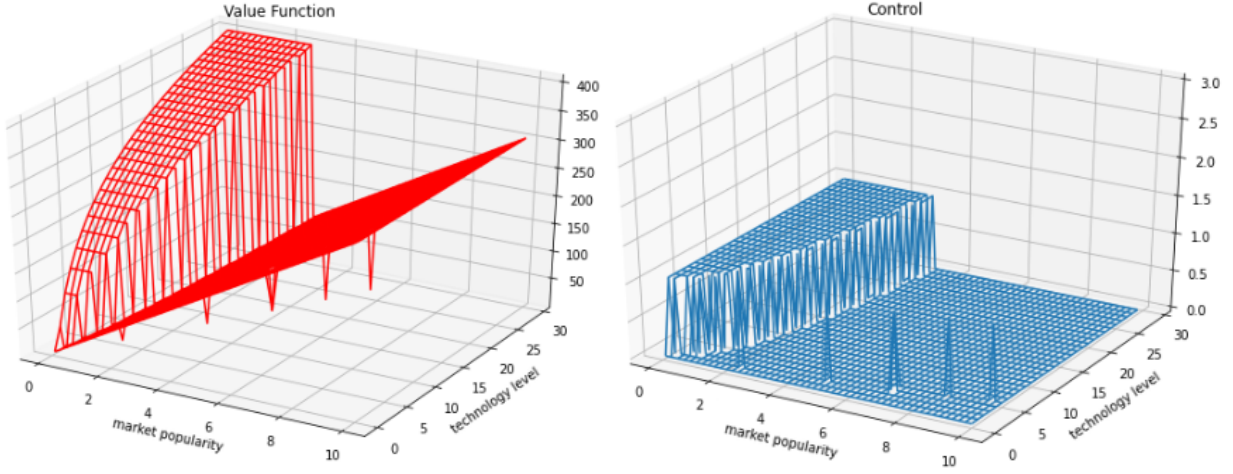


Figure 4.12: Control and Value Function for $\pi_1 = z_1 \log(1 + \theta)$

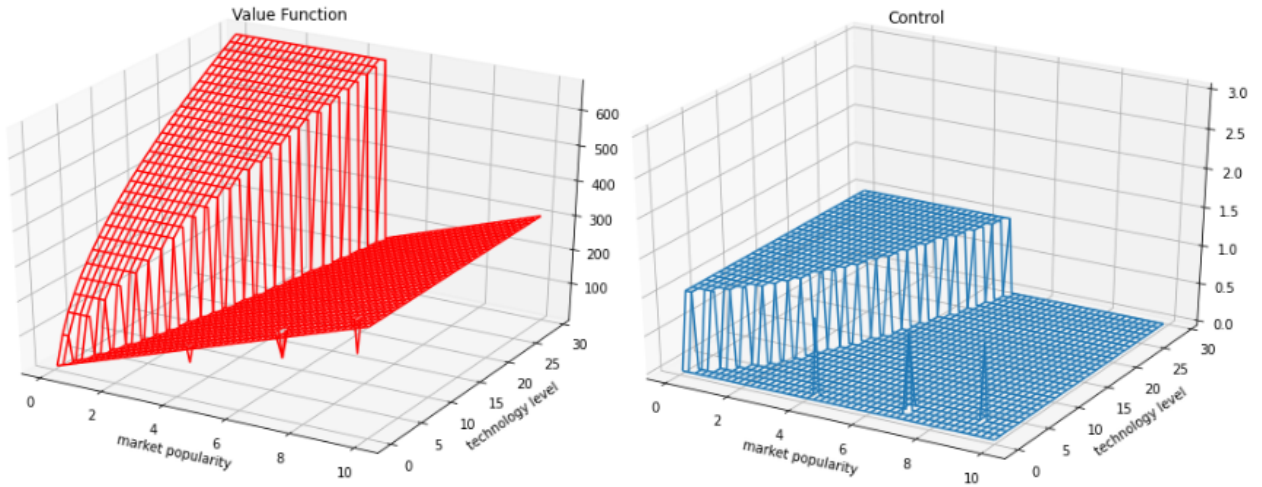


Figure 4.13: Control and Value Function for $\pi_1 = z_1 \sqrt[3]{\theta}$

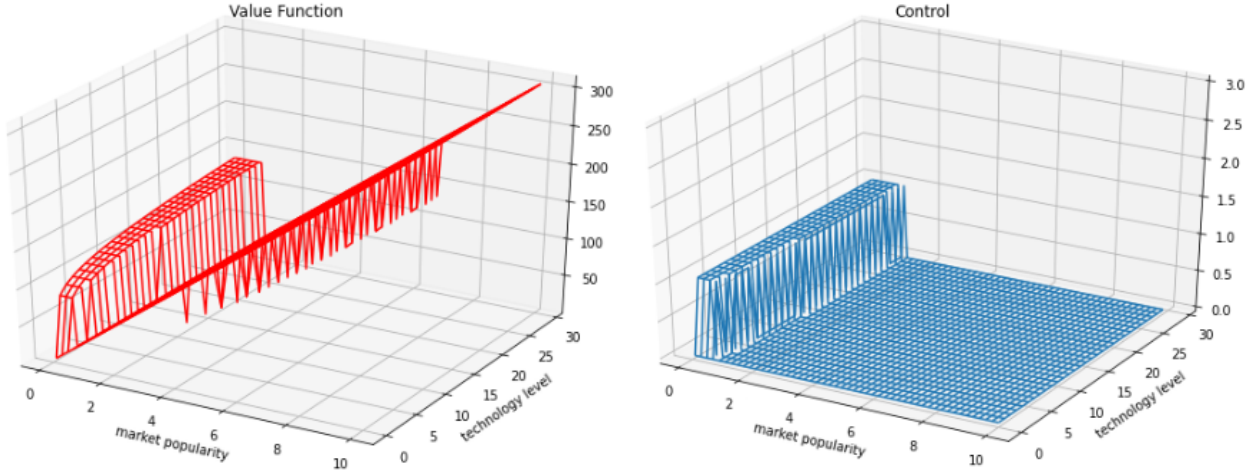
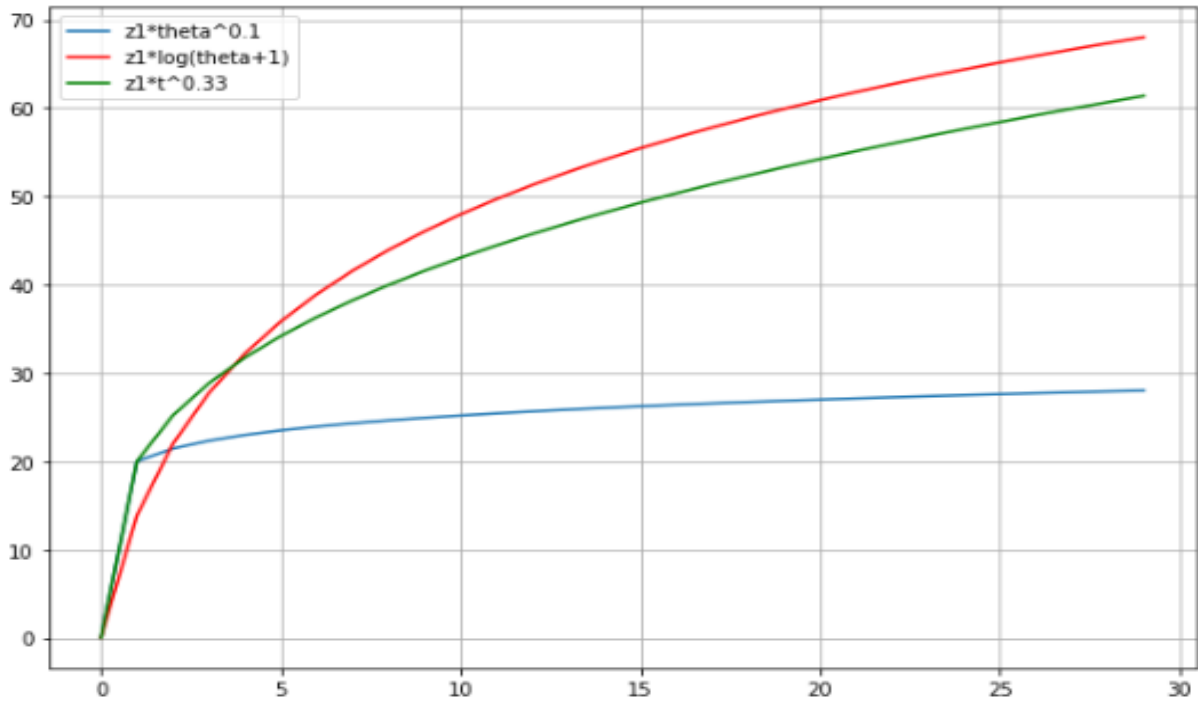
Figure 4.14: Control and Value Function for $\pi_1 = z_1 \sqrt[10]{\theta}$ 

Figure 4.15: Profit flow function

We can see from 4.15 that $z_1 \sqrt[10]{\theta} < z_1 \sqrt[3]{\theta}, \forall \theta$, and from figures 4.14 and 4.13 that the stopping region in 4.14, (with a profit flow function $z_1 \sqrt[10]{\theta}$) is smaller than the stopping region in 4.13 (with a profit flow function $z_1 \sqrt[3]{\theta}$) which corresponds to our previous suppositions.

4.4 Conclusion

To conclude, in this last chapter, we took a look on the implementation of the Howard algorithm for this model and how our implementation differs from previous studied ones in [5] since our model uses two variables, hence the use of the Sylvester Equation. We also explained and showed the results in a general way in order to understand them. Then we proceeded with a sensitivity analysis to get a deeper understanding of our results and see how rigorous they are. The results are generally quite satisfactory. The only downside to this algorithm is the relatively long run-time.

General Conclusion and Perspectives

In this project, we studied the optimal time for a brick-and-mortar retailer to switch to an online retailing method, taking into consideration two state variables: market popularity and the technology level of data usage. For this study we used an optimal stopping time approach and we used Howard Algorithm to resolve it.

The first chapter of this work was fully dedicated to build the general framework. First we presented the host organization of this study. Then we stated the motivations behind this work why it could be helpful for retailers. Finally we reviewed some of the prior works and publications that helped the elaboration of this project.

In the second chapter, we presented the different theoretical tools used for our purpose. These tools included a variety of domains notably: dynamic programming, optimal control theory and optimization.

In the third chapter, we showed the model which was originally made in [3]. We presented the different parameters, functions and equations that governed this model. In this chapter, we also spoke about the added value that was given by this work and how it differs from the original model.

Lastly, we tried to explain the implementation of this work and the results given by it. To have a better understanding of these results, we conducted a sensitivity analysis in order to see the contribution of each parameter to the model and how conform is it to real life.

A potential perspective for this work, is to see the case of a stochastic market where it would be harder to determine the optimal stopping time because of the uncertainty of the market. We can also widen our options by considering the case where the retailer's decision is to add an online platform to his brick and mortar store and not simply switching.

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